UNIT 3 · STANDARD 6.RP.3

Use Ratio Reasoning

Lesson 3-6

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

2/3 = 8/12

Proportion

A math sentence saying two ratios are equal.

2/3 = 8/12 → 2×12 = 24 and 3×8 = 24 ✓

Cross-multiply

Multiplying across two ratios to check if they are equal.

5:8 × 3 → 15:24

Scale

To multiply or divide both parts of a ratio by the same number.

1/2 and 3/6 and 5/10 all equal the same amount — half

Equivalent

Having the same value.

60 miles in 3 hours → 20 miles per 1 hour

Unit rate

A rate for just 1 of something. You find it by dividing.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can use ratio reasoning to solve problems with proportions and scaling.

1️⃣ Notice

Chef Academy received a huge catering order — a banquet for 120 guests!

2️⃣ Key idea

I can use ratio reasoning to solve problems with proportions and scaling.

3️⃣ Words I'll use

Proportion **Cross-multiply** **Scale** **Equivalent** **Unit rate**

🤔 Think About It

- How many times bigger is 120 compared to 8?
- What operation would help you scale both ingredients?
- What stays the same about the recipe even when the amounts change?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Reyes's recipe serves 8 people, but the banquet has 120 guests. How many times bigger is 120 than 8, and how does that scale factor help you adjust the tomatoes and mozzarella?

Level 1 support

Start here: Find the scale factor (120 divided by 8 = 15), then multiply each ingredient by it.

💡 **Need a hint?**

Sentence starters (optional)

120 guests is ____ times bigger than 8 because 120 divided by 8 equals ____.

120 invitados es ____ veces más que 8 porque 120 entre 8 es ____.

So I scale by ____, which keeps the recipe in ____.

Entonces escalo por ____, lo que mantiene la receta en ____.

Word bank: proportion | scale | equivalent | unit rate | ratio

Listen for: *Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).*

Level 2

What stays the same about the recipe even when all the amounts get bigger? Justify why scaling keeps the flavor the same.

- The ____ between tomatoes and mozzarella stays the same because ____.
- Scaling keeps the flavor because ____.

EXPLORE

You sorted ratio pairs into equivalent and not equivalent. How does ratio reasoning, like cross-multiplying, prove that 4:7 and 12:21 are equivalent?

Level 1 support

Start here: Cross-multiplying gives equal products when two ratios form a true proportion.

💡 **Need a hint?**

Sentence starters (optional)

I reasoned that 4:7 and 12:21 are ___ because ___.

Razoné que 4:7 y 12:21 son ___ porque ___.

Cross-multiplying gives ___ and ___, which are ___.

La multiplicación cruzada da ___ y ___, que son ___.

Word bank: **proportion** **equivalent** **scale** **cross-multiply** **ratio**

Listen for: *Students show $4 \times 21 = 84$ and $7 \times 12 = 84$ (or that 4:7 scales by 3 to 12:21), proving the ratios are equivalent.*

Level 2

Two ratios cross-multiply to UNequal products. Explain what that tells you and give an example you tested.

- If the cross products are not equal, then ___.
- An example I tested is ___ because ___.

CONNECT

Where might ratio reasoning help you make a smart choice in real life, like shopping or cooking?

Level 1 support

Start here: Scaling recipes, comparing prices, and mixing ingredients all use ratio reasoning.

💡 **Need a hint?**

Sentence starters (optional)

Ratio reasoning helps me when ____.

El razonamiento con razones me ayuda cuando ____.

I used the ratio to decide ____.

Usé la razón para decidir ____.

Word bank: **proportion** **scale** **equivalent** **unit rate** **ratio**

Listen for: *Students name a real use (scaling a recipe, comparing deals, mixing paint or fuel) and explain how keeping the ratio equivalent guides the decision.*

Level 2

Describe a situation where ignoring the ratio would cause a real problem, and explain how proportional reasoning fixes it.

- Ignoring the ratio would cause a problem when ____.
- Proportional reasoning fixes it by ____.

Guided Examples

Example 1

A recipe uses 4 cups of broth for every 10 servings. How many cups of broth are needed for 30 servings?

Solution: Scale factor: $30 \div 10 = 3$. Multiply broth by 3: $4 \times 3 = 12$ cups.

Answer: A. 12

Example 2

Are the ratios 6:9 and 2:3 equivalent?

Solution: Divide 6:9 by 3 to get 2:3. Since both ratios simplify to 2:3, they are equivalent.

Answer: A. Yes, both simplify to 2:3

Example 3

If 3 pounds of apples cost $6, how much do 7 pounds cost?

Solution: Unit rate: $\$6 \div 3 = \$2$ per pound. For 7 pounds: $\$2 \times 7 = \$14$.

Answer: A. $14

Write About the Math

The Writing Revolution

I can explain my reasoning using the words proportion, scale, equivalent, and unit rate.

1. Kernel Sentence

subject + verb

Model: Scale is to multiply or divide both parts of a ratio by the same number.
Escalar es multiplicar o dividir ambas partes de una razón por el mismo número.

Write a kernel sentence about scale. Use a subject and a verb.
Escribe una oración base sobre escalar. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Scale matters in math
Escalar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Scale matters in math because ____.
Escalar importa en matemáticas porque ____.

___

but
pero

Scale matters in math, but ____.
Escalar importa en matemáticas, pero ____.

___

so
entonces

Scale matters in math, so ____.
Escalar importa en matemáticas, entonces ____.

___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about scale.
Di un hecho verdadero sobre scale.

Scale ____.

Question
Pregunta

Ask a question about scale.
Haz una pregunta sobre scale.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about scale.
Muestra entusiasmo sobre scale.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with scale.
Dile a un compañero qué hacer con scale.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

120 guests is ____ times bigger than 8 because 120 divided by 8 equals ____.
120 invitados es ____ veces más que 8 porque 120 entre 8 es ____.

So I scale by ____, which keeps the recipe in ____.
Entonces escalo por ____, lo que mantiene la receta en ____.

I reasoned that 4:7 and 12:21 are ____ because ____.
Razoné que 4:7 y 12:21 son ____ porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A chef uses the proportion 3/7 = x/28 to scale a recipe. What is x?

A. 12

B. 14

C. 9

D. 21

Show your work:

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2. A recipe serves 6 people and uses 4 cups of rice. You need to serve 15 people. Show two different methods to find how much rice you need: (1) using a scale factor and (2) using a unit rate.

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Find Priya's Mistake — find the error, then write the correct reasoning.

Show your work:

Reflect — Exit Ticket

A recipe calls for 6 cups of flour for every 15 cookies. How many cups of flour are needed to make 45 cookies?

A. 18

B. 12

C. 24

D. 21

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 12 *— Scale factor: 28 ÷ 7 = 4. Multiply: 3 × 4 = 12. So x = 12.*
2. **Try It 2:** Method 1 (scale factor): 15 ÷ 6 = 2.5. Multiply rice by 2.5: 4 × 2.5 = 10 cups. Method 2 (unit rate): 4 ÷ 6 = 2/3 cup per person. For 15 people: 2/3 × 15 = 10 cups. Both methods give 10 cups of rice.
3. **Exit Ticket:** A. 18 *— Scale factor: 45 ÷ 15 = 3. Multiply flour by 3: 6 × 3 = 18 cups of flour.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Scale is to multiply or divide both parts of a ratio by the same number.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).